

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**End Semester Examination – Summer 2019****Course: B. Tech in Computer Engineering****Sem: III****Subject Name: Data Structures****Subject Code: BTCOC303****Max Marks:60****Date: 30/05/2019****Duration: 3 Hr.****Instructions to the Students:**

1. Solve **ANY FIVE** questions out of the following.
2. The level question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		
A) What is Data Structure? Explain the various characteristics of an algorithm		6
B) What is time complexity? Compute the frequency count for : for i : = 1 to n for j : = i + 1 to n for k : = j + 1 to n for l : = k + 1 to n x = x + 1;		6
C) What is an algorithm? Write an algorithm to find Greatest common divisor (GCD).		6
Q.2 Solve the following.		
A) Write a “C” code to find the transpose of a sparse matrix stored in this way.		6
B) Using linear probing insert the following values in hash table of size 10. Elements are 28, 55, 71, 67, 11, 10, 90, 44.		6
Q. 3 Solve the following.		
A) Explain sequential search. Write an algorithm for sequential search.		4
B) What is skip list? Give its representation .Write an algorithm to insert new item (k,e) in the skip list S.		8

Q.4 Solve the following.

- A) Write a program in C to create a singly linked list and perform the following operations I) Insert into list II) Search for data III) Delete from list **6**
- B) Construct algorithm for following operations on a Doubly Linked List 1) CREATE AT END 2) DELETE AT START 3) TRAVERSE **6**

Q.5 Solve the following.

- A) With the help of suitable example, explain following operation, Enqueue and Dequeue and traverse operation of circular queue **6**
- B) Convert the $A*B+C/D$ expression into postfix using stack **6**

Q.6 Solve the following.

- A) Explain breadth first search technique for graph traversal. **6**
- B) What is a Binary Tree. Explain inorder and postorder traversals with example **6**

*** End ***